

CHAPTER I

INTRODUCTION

Cities keep getting more crowded, and parking is becoming a real daily problem. A 2023 report from the International Transport Forum says drivers in big cities lose about seventeen hours a year just hunting for a parking spot. That burns extra fuel, causes frustration, and adds pollution. Some cities like San Francisco, Tokyo, and Amsterdam have already set up smart parking systems that use cameras and AI. These systems tell drivers where open slots are, and a few cities have cut down circling traffic by around thirty percent. So smart parking isn't just a sci-fi idea anymore. It's real, and it works.

Southeast Asia is starting to see the same thing happen. Singapore has a program called Smart Nation that uses cameras to point drivers toward empty parking spaces. Malaysia and Thailand are also trying out small projects where sensors and AI work together to watch over parking lots. These examples show that smart parking really works in different countries and situations. So technology can clearly help with the problem of finding a parking spot. But the real question is whether the Philippines has caught up yet.

The Department of Information and Communications Technology (DICT) has listed intelligent transportation systems, including smart parking, as a national priority. The Land Transportation Office says there are now over fifteen million registered vehicles nationwide, growing about six percent each year. However, most smart parking systems are still found only in a few high-end malls in Metro Manila. In provincial cities like Bacolod, parking is still managed the old-fashioned way. Guards walk around to visually check slots, drivers keep circling, and managers have no real data to use. That old system wastes time and fuel. It frustrates drivers, tires out guards, and offers no records for planning.

Bacolod City is growing fast, and parking gets worse each year. The City Planning and Development Office reported that registered vehicles go up by eight percent every year. Some areas get really congested during peak hours – like Lacson Street, downtown, and the Burgos-Rizal area. We chose one pilot site as our example: Burgos,

on BBB Street, because we saw the problem ourselves. From March 2026, we timed drivers taking five to eight minutes just to find an open spot. Guards had no help – they just used their eyes and walkie-talkies. The CCTV cameras worked fine, but they only recorded for security, not for parking.

That was the gap we wanted to fill with our SmartPark project. International studies have already proven that YOLOv8 and Deep SORT work well for parking monitoring. But no one had applied that combination in Bacolod City or any similar Philippine secondary city. Some local projects used expensive per-slot sensors that cost too much for larger lots. Others used older image processing methods that were not very accurate in real conditions. None of them integrated detection, tracking, slot polygons, and a web dashboard into one complete system. And none of them respected the Data Privacy Act while doing so. That missing piece became our main goal.

So we built SmartPark to solve these problems using existing CCTV cameras. It takes the live feeds from the cameras already installed at our chosen site. Then it runs those feeds through YOLOv8 for vehicle detection and Deep SORT for tracking. After that, the system checks which slot polygons are overlapped by vehicles and marks those slots as occupied. All the information shows up on a simple web dashboard that guards can open on any browser. We pilot-tested SmartPark in Bacolod, measured its performance, and proved that it works. This is our contribution to SDG 9 for industry and innovation, and to SDG 11 for sustainable cities and communities.

Project Context

We spent time at our example parking site–Burgos, on BBB Street in Bacolod City–to see how parking works in real conditions. We talked to the guards, watched how drivers moved around, and wrote down what we saw. We found that guards have no central screen to check open slots. They walk around or shout on walkie-talkie to ask other guards, and sometimes that leads to wrong directions. There is also no system that tracks occupancy data. The administrator usually finds out the lot is full only when cars start lining up on the road. Entry and exit logs, if they bother to write them, go on paper

and rarely get used for planning. The CCTV cameras already cover the whole lot, but they only connect to a DVR for security recording. No one uses them to help with parking.

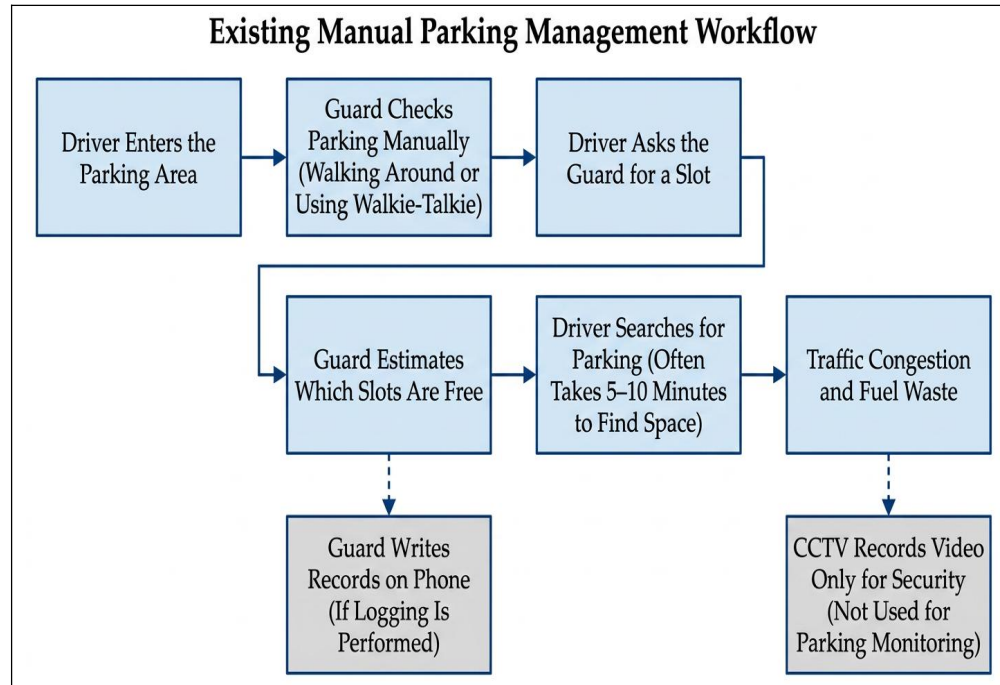


Figure 1. Workflow of the Existing Manual Parking Management System.

The project proponents designed and developed a web-based application to address the problems experienced by the beneficiaries in the manual-based transaction. These are the following:

1. Guards had no centralized screen or real-time reference for slot availability, causing delayed checking and occasional inaccurate directions to drivers.
2. There was no system that collected parking occupancy data, so administrators only confirmed that the lot was full when vehicles started lining up.
3. Entry and exit logging, when done, was recorded manually on paper or informal notes, and these records were rarely used for reporting or planning.

4. CCTV cameras were already installed and functional but were used only for security recording and not utilized for parking monitoring or operational decision-making.

Purpose and Description

We built SmartPark as a useful system to fix the parking problems we saw at our chosen site. It is a camera-based system that watches parking lots smartly. It automatically detects vehicles, tracks them, checks which slots are taken, and shows everything on a web dashboard. The system pulls live video from the CCTV cameras that were already there. YOLOv8 spots the vehicles and draws boxes around them. Then Deep SORT gives each vehicle a unique ID so the system can follow it as it moves from one frame to the next. After that, we compare where the vehicles are with the slot polygons we drew earlier. If a vehicle's box touches a polygon, we mark that slot as occupied. If not, it stays available. This all happens almost in real time. The dashboard refreshes every few seconds, so guards and admins can see the current parking status on dashboard.

Objectives

The goal of this study is to build a camera-based parking monitoring system that uses AI to detect vehicles, paired with a web application, so parking management becomes faster and less subject on manual work. Rather than having guards walk around and shout, the system will provide real-time updates and automatically collect data. Specifically, we aimed to do the following:

1. Create a vehicle detector using YOLOv8 that works with the CCTV cameras already installed at our chosen site. The detector must spot vehicles accurately under real conditions—daytime, night with decent lighting, and normal traffic flow.

2. Set up automatic logging if when each vehicle enters and leaves. From those timestamps, the system will figure out the parking duration and save everything in a database for next reporting and analysis.
3. Build a web dashboard that shows live slot availability, how many vehicles are inside, and a warning when the lot is nearly full. Administrators and parking guards must also be able to download past reports (PDF or Excel) without extra steps.
4. Test how accurate the detection is, whether the tracking works well, and if actual users find the system helpful. We used a simple three-point scale (Good, Acceptable, Needs Improvement) and asked parking guards and site administrators to rate SmartPark based on their daily experience.

Significance of the Study

SmartPark was built to help people who deal with parking problems every day. Based on what we saw at our example chosen site in Bacolod City, we identified seven groups that will benefit from this system.

Infrastructure Administrators and Facility Managers.

They get a clear, data-driven view of the parking lot. They can see which hours are busiest and how long people usually stay. That helps them decide when to add more staff or change the layout. Real-time alerts also tell them the moment the lot gets full.

Security Guards and Parking Attendants.

Instead of walking around under the sun or shouting into walkie-talkies, they simply look at the dashboard. It saves their energy and gives them accurate info. They can point drivers to an empty slot without second-guessing.

Motorists and the General Public.

Drivers no longer need to circle the lot for five to ten minutes. That means less frustration, less wasted fuel, and cleaner air. In the future, we could even put a public screen outside so drivers see open slots before entering.

Establishment Owners and Operators.

A smoother parking experience keeps customers happy. Happy customers tend to stay longer and come back. That means more sales and repeat business.

Bacolod City Local Government Unit.

This project proves that a smart parking system does not have to be expensive. The city can take our low-cost model and use it in other public or commercial parking areas to help ease traffic around the city.

Future Researchers.

They can use SmartPark as a starting point. For example, they could add license plate recognition, build a mobile app for drivers, or predict how full the lot will be at different times of day.

United Nations Sustainable Development Goals.

SmartPark supports SDG 9 (Industry, Innovation, Infrastructure) by putting AI to work solving a real everyday problem. It also supports SDG 11 (Sustainable Cities and Communities) by reducing traffic, lowering vehicle emissions, and making urban travel a little bit better.

Scope and Limitations

The general purpose of this study is to design, develop, and test a camera-based parking monitoring system that uses AI to detect vehicles and a web dashboard to show real-time occupancy. We built and tested SmartPark at one selected parking facility in Bacolod City. The target users are administrators, facility managers, security guards, and parking attendants at that site. The development and testing took place this school year 2026-2027. The system detects vehicles in real time using YOLOv8 from live CCTV feeds. It also tracks vehicles using Deep SORT to avoid double counting. Occupancy is checked using slot polygons that the admin draws on the camera view. The web dashboard shows live slot status, past analytics, and alerts, and admin can download reports in PDF or Excel.

We also know that some things are outside our control. How well the system detects vehicles depends on the camera's quality, resolution, angle, and where it is placed. If a camera runs below 720p or has a blocked view, the system will not perform as well. Bad weather, like heavy rain or fog, and very dim lighting will also make detection harder. SmartPark works best during the day or at night with good artificial lights. It cannot handle total darkness. The web dashboard also needs a steady local network connection between the server and the devices viewing it. Short network drops are okay, but if the connection stays down for too long, updates will be delayed. We only tested our prototype at one parking site in Bacolod City. That means the results might not be exactly the same if someone tries it at a different location. Lastly, we had to rely on the facility's management for camera access, network access, and on-site setup. Any limits they placed on us were simply beyond what we could change.

Definition of Terms

Below are the key terms we used throughout this study. We defined each one the way it applies specifically to SmartPark. The terms are arranged alphabetically.

Bacolod City.

The highly urbanized capital of Negros Occidental, Philippines. This is where we set up and tested SmartPark.

Bounding Box.

The rectangle that YOLOv8 draws around a detected vehicle. It shows where the vehicle is and roughly how big it is.

CCTV (Closed-Circuit Television).

The cameras already installed at our chosen site. We used their video feeds as input to the system.

Computer Vision.

A branch of AI that lets machines understand and pull information from images and videos.

Dashboard.

The web-based screen that shows real-time occupancy, past analytics, and reports to administrators and guards.

Deep SORT.

Short for "Simple Online and Realtime Tracking with a Deep Association Metric." This is the algorithm we used to track vehicles as they move from one video frame to the next.

Detection Accuracy.

How well YOLOv8 correctly spots vehicles. We measured this using mean average precision, or mAP.

Mean Average Precision (mAP).

A standard way to measure object detection performance. It averages the precision values across all vehicle types at a given overlap threshold.

Occupancy.

The status of a parking slot. We mark it "occupied" if a tracked vehicle's bounding box touches the slot polygon, and "available" if not.

Parking Duration.

The time between a vehicle's entry and exit events. The system calculates this automatically from the timestamps.

Slot- Polygon Mapping.

A method where an admin draws polygonal shapes on the camera view to mark each parking slot. The system then checks if a vehicle overlaps those shapes to decide occupancy.

Tracking.

Keeping the same ID on a vehicle as it moves across video frames. We used Deep SORT to do this.

Vehicle Detection.

Finding and locating vehicles in each video frame using YOLOv8.

Web Application.

The part of SmartPark that runs in a browser. It connects the AI backend to the people using the system.

CHAPTER II

REVIEW OF RELATED LITERATURE/STUDIES/SYSTEMS

This chapter presents the related literature, studies, and systems we reviewed for SmartPark. Our project is a camera-based parking monitoring system that uses YOLOv8 for vehicle detection and Deep SORT for tracking, built for a selected site in Bacolod City. We needed to understand what other researchers and developers have already done in this area. So we looked at both foreign and local sources. We focused on smart parking systems, computer vision, vehicle detection, multi-object tracking, CCTV-based monitoring, and web-based parking management. These topics directly relate to what we built. The review helped us see what methods work well and where gaps still exist. It also gave us a solid technical and practical foundation for our own project.

Foreign Related Literature

Smart Parking Systems.

Smart parking systems use technology to help drivers and parking staff monitor slot availability. Traditional manual methods where guards walk around or rely on memory are slow and often wrong. Smart systems replace this with sensors, cameras, and AI to do the job automatically. A 2023 review published in Transportation Research Procedia looked at 124 research papers on smart parking systems. The authors found that wireless sensor networks have become popular because they are cheaper to install than running wires everywhere. They also noted that computer vision systems need fewer sensors since one camera can watch many slots, though setting up the cameras themselves costs more upfront. This review is helpful because it confirms that our approach using cameras instead of per-slot sensors is a recognised direction in smart parking research.

Camera-Based Parking Occupancy Detection.

Instead of burying a sensor under every single slot, you can point one camera at a whole row of slots. This works out cheaper, especially when a parking lot already has CCTV cameras installed for security. A systematic review from 2022, published in *Expert Systems with Applications*, examined all the research on computer vision methods for parking lot management. The authors looked at dozens of studies that used deep learning to detect whether a slot is empty or occupied from video feeds. Their review confirms that camera-based methods are practical and widely studied, even under different lighting and weather conditions. This reinforces our decision to use the existing CCTV feeds at our Bacolod site instead of buying new hardware.

YOLOv8 Vehicle Detection.

YOLO stands for You Only Look Once, and it is a real-time object detection model that is both fast and reasonably accurate. YOLOv8 is the latest version from Ultralytics, and many researchers have found it works well for spotting vehicles in parking lots. A 2024 conference paper from the IEEE Beyond Technology Summit tested YOLOv8 specifically for real-time parking space detection. The authors trained the model on recorded parking videos and reported a detection accuracy of up to 99.1% under various lighting conditions. This is important for our project because high accuracy means fewer false reports of empty or occupied slots. It also shows that YOLOv8 can handle the kind of daytime and night-time conditions we expect at our site in Bacolod.

Deep SORT Vehicle Tracking.

Detection alone is not enough. Without tracking, the same vehicle appearing in multiple video frames would be counted many times, messing up occupancy and duration calculations. Deep SORT assigns a unique ID to each vehicle and follows it across frames using a combination of motion and appearance features. A 2022 paper published in *Sensors* proposed an improved Deep SORT algorithm for open-space vehicle positioning. The authors used it to track vehicles and even their license plates across consecutive frames, and they found that the tracking accuracy was high enough for real-world parking management. This is exactly why we paired YOLOv8 with Deep

SORT in SmartPark – to track vehicles reliably from the moment they enter until they leave.

Web-Based Monitoring Dashboards.

A dashboard that runs in a web browser is practical. No one needs to install special software; guards and administrators can just open a page on a desktop, tablet, or even a phone. A 2023 book chapter titled Intelligent Dashboards for Car Parking Flow Monitoring discussed how to build such dashboards using online analytical processing (OLAP). The authors built a system that extracts data from parking sensors, stores it in a data warehouse, and then displays key indicators such as which parking areas are busiest and when. This supports our decision to build a browser-based dashboard for SmartPark, where users can see live slot status, historical reports, and alerts without any complicated setup.

Local Related Literature

Intelligent Transportation Systems in the Philippines.

The Philippine government has been pushing for digital transformation in transportation. The Department of Transportation (DOTr) and the Department of Information and Communications Technology (DICT) signed a memorandum of understanding in April 2023 to work together on transport infrastructure projects through e-governance. Under this agreement, the DICT helps the DOTr identify and develop new systems and applications to deliver better public service. A market report also notes that as part of urban mobility reforms, the DOTr is integrating smart parking into the National Transport Policy. National guidelines encourage the use of technology for traffic and parking management. This means our SmartPark project is aligned with current government directions, even on a local level in Bacolod.

Parking and Traffic Congestion in Philippine Urban Areas.

In cities like Manila, Cebu, and Bacolod, drivers waste a lot of time looking for parking. Manual systems make the problem worse because guards cannot give accurate real-time information. According to the Bacolod Traffic Authority Office (BTAO), the city has been experiencing a "severe strain" on its transport capacity, threatening its

livability. As one of the most highly urbanized cities in the country, Bacolod faces heavy traffic and congestion on a daily basis. In fact, it was noted in 2024 that radar-assisted traffic lights were installed at five major intersections to help improve flow, partly due to parking-related issues. These conditions match what we observed at our selected site on Burgos Street, where drivers routinely circle for several minutes before finding a spot.

CCTV Infrastructure in Local Establishments.

Many malls, schools, hospitals, and offices in the Philippines already have CCTV cameras installed. However, most establishments use them only for security recording, not for operational intelligence like parking monitoring. Local studies show that CCTV surveillance is actually the most favored safety feature among residents when choosing a parking facility, underscoring the importance of video security. For example, a recent study developed a smart parking system for Centro Escolar University – Malolos using a webcam and an existing CCTV camera alongside OpenCV for image processing. The researchers found that camera-based solutions are practical and ready for deployment. This supports our idea of reusing CCTV feeds for parking intelligence rather than buying expensive per-slot sensors.

Data Privacy and Responsible Use of Video Analytics.

Republic Act No. 10173, also known as the Data Privacy Act of 2012, requires everyone to handle personal information responsibly. The law states that it is the policy of the State to protect the fundamental human right to privacy while ensuring the free flow of information to promote innovation and growth. Because a camera-based parking system captures video, we must be careful not to violate this law. Our system avoids facial recognition and does not identify drivers or passengers. We only collect vehicle data and slot occupancy information. This responsible approach respects privacy while still delivering the benefits of AI-based parking monitoring.

Smart City Development in Bacolod City.

Bacolod is growing, and with growth comes more vehicles and greater parking demand. The city government has taken concrete steps toward becoming a smarter city. In March 2025, Bacolod entered into a 10-year public-private partnership to develop the

"Bacolod Super City Project," which includes a centralized command center for monitoring traffic and emergency responses. The plan also includes installing intelligent streetlamp poles equipped with video surveillance and analytics in about 150 locations. A separate initiative introduced radar-assisted traffic lights in key intersections to dynamically adjust timings based on real-time traffic volume. These developments show that the city is ready for technology-based solutions. We believe SmartPark can be a small step toward making Bacolod a smarter city by showing how existing CCTV cameras can be used for parking intelligence in a low-cost and scalable way.

Foreign Related Studies and/or Systems

Deep Learning for Decentralized Parking Lot Occupancy Detection (Amato et al., 2016).

This study was the first we encountered that directly inspired our approach. Amato and his team proposed a decentralized parking occupancy detection system using a deep Convolutional Neural Network (CNN) designed to run directly on smart cameras. They showed that you do not need a central server for every decision; the camera itself can tell if a slot is free or taken. What impressed us was how they avoided per-slot sensors and relied purely on image analysis. Their CNN was compact enough to run on limited hardware, yet its accuracy was comparable to the much larger AlexNet model. This proved that camera-only solutions are not just possible, but practical. For SmartPark, their work gave us confidence that reusing CCTV feeds without adding expensive hardware is a valid and efficient path to follow.

CMCA-YOLO for Parking Lot Surveillance Imagery (Zhao et al., 2024).

This is a very recent study that pushed YOLO-based detection further for parking environments. Zhao and his colleagues faced a common problem in parking lots: vehicles far from the camera appear small and often overlap with each other, making them hard to detect. They proposed a new model called CMCA-YOLO, which stands for Criss-cross and Multi-spectral Channel Attention. They added special attention modules that help the model focus on the right pixels even when targets are tiny or overlapping. On their own dataset of 4,502 parking lot images, CMCA-YOLO achieved an mAP@0.5 score of

0.895, outperforming standard YOLO. For us, this study confirmed that YOLO is not just an academic toy; it is being actively improved for real-world parking challenges like varying angles, distances, and lighting. It reinforced our decision to use YOLOv8 as the core of SmartPark.

Edge AI-Based Vehicle Tracking for Smart Parking (Doan et al., 2023).

This study addressed a practical concern we had: does the AI processing have to happen on an expensive central computer? Doan and his team explored a different setup. They proposed running vehicle detection and tracking on edge devices – small, affordable computers placed right next to the cameras. Their system used a diffusion model and an information potential field to navigate vehicles to empty spaces. The key finding for us is that edge processing reduces latency because video frames do not have to travel far to be analyzed. This means updates to the dashboard can be nearly instant. While SmartPark currently runs on a central server, Doan's work showed us a possible upgrade path. In the future, we could install an edge device at the parking gate to keep the system running even if the main network is slow or down. This makes the whole monitoring setup more reliable.

Smart Real-Time Parking Control and Monitoring System (Elfaki et al., 2023).

Elfaki and his co-authors built and field-tested a complete smart parking system, which they published in the journal *Sensors*. Their work covered exactly the three pillars we care about: real-time monitoring, data communication, and user interfaces. They measured not just detection accuracy but also transmission delay and processing time – practical metrics that matter when you are deploying a system in a real parking lot. What we appreciated most is that they did not stop at the lab. They took their system into the field and measured how it performed under real conditions. This directly matches our own approach. Like Elfaki, we are building a complete system that connects detection, tracking, a database, and a dashboard. His work gave us a reference for what metrics to track during our own evaluation at the Bacolod site.

Artificial Intelligence- Enabled Smart Parking System (Singh et al., 2024).

This study took a broad view, looking not just at detection but at the entire ecosystem of AI, IoT, and sensor technologies for parking. The authors explored how AI can work alongside Internet of Things devices – like smart sensors and cameras – to collect real-time data, process it, and deliver parking availability updates to drivers via mobile apps or display boards. Their main conclusion was that AI-based parking systems significantly improve efficiency compared to traditional methods. They also noted that such systems reduce traffic congestion and improve the parking experience for drivers. For SmartPark, this study helped us see the bigger picture: we are not just building a detector. We are building a system that should eventually integrate with other urban technologies. It reaffirmed that a camera-based AI system is not a niche experiment but a mainstream direction for smart cities.

Local Related Studies and/or Systems

Sensor-Based Parking Monitoring Systems in Philippine Institutions.

Several local undergraduate theses have built parking monitoring systems using ultrasonic or infrared sensors placed in every slot. For instance, a study from Bestlink College of the Philippines in 2026 developed a smart parking system using IR proximity sensors to manage entry and exit points, with an I2C display for user guidance and an Arduino Uno as the central controller. Another project from the University of Mindanao used ultrasonic sensors to track vehicle entry and exit, displaying real-time availability on a mobile app. These projects prove that sensor-based monitoring works. However, for a large parking area, installing one sensor per slot becomes expensive and difficult to scale. That inefficiency led us to pursue a camera-based approach for SmartPark, where a single CCTV can monitor many slots at once.

CCTV-Based Traffic Monitoring in Philippine Cities.

Local government units in Metro Manila, Cebu, and Davao have long used CCTV cameras for traffic monitoring, but almost always on roads rather than inside parking lots. For instance, the Metro Cebu Bridge Management Board inaugurated a ₱9.5-million command center in March 2025, featuring 39 solar-powered wireless CsisCCTV cameras

with AI capabilities like vehicle detection and license plate recognition. Similarly, Dumaguete City has started installing a Smart CCTV Traffic Lights System with vehicle-detection cameras at key intersections. These initiatives confirm that LGUs trust CCTV feeds for real-time analysis. For SmartPark, we simply transfer that same logic from roads to parking slots, using the existing CCTV infrastructure of a single Bacolod establishment.

Web-Based Parking Information Systems.

Some local capstone projects already include web dashboards for parking, but a common drawback is that they require manual entry of slot availability. One exception is the "SPARK" system from UM Tagum College, which uses computer vision to automatically detect empty slots and display them on a web-based app. Their testing showed it was able to accurately detect and track parking space availability, and the web app was user-friendly. Another study from the Nueva Ecija University of Science and Technology developed a web-based parking lot management system using HTML, CSS, JavaScript, and MySQL. That system focuses on monitoring entries, exits, and income. These studies show that web dashboards are effective when paired with automatic sensors or cameras. SmartPark improves on them by adding automatic tracking (Deep SORT) and downloadable historical reports, reducing the need for guard intervention.

Manual Parking Management in Bacolod City Establishments.

From our own site visits along Burgos Street in Bacolod, we observed firsthand that most parking management is still done by hand. Guards use hand signals, walkie-talkies, and paper logs to record entries and exits. Drivers routinely circle for five to ten minutes because no real-time information is available. These observations match the findings of a 2007 thesis from Central Philippine University, which noted that manual parking systems on campus relied on badge swipe devices and lacked real-time monitoring. The contrast between what we saw on site and what the literature says is possible with AI gave us the strongest motivation to build SmartPark.

Existing CCTV Infrastructure at Selected Bacolod City Site.

Our chosen parking facility along Burgos Street already had functional CCTV cameras covering the entire lot. However, those cameras were connected only to a DVR for security recording; their video feeds were never used for parking intelligence. This situation is common in many Philippine establishments, where CCTV infrastructure remains under-utilised. A 2023 DLSU senior high school conference paper developed a smart parking reservation system called "Parkchers" that integrated CCTV camera-based software to test detection under various lighting scenarios. Their results showed that precision and recall exceeded 96% by mid-afternoon, though nighttime low light remained a challenge. Their success gave us confidence that the existing CCTV at our Bacolod site could be repurposed in the same way, turning a passive security tool into an active parking management system for SmartPark.

Synthesis

After reviewing all the related literature and studies, we saw a clear pattern. The Philippines is slowly moving toward smart cities, but parking management in most provincial areas like Bacolod remains stuck in the old way. Many establishments already have CCTV cameras, yet they only use them for security recording, not for monitoring parking. Local smart parking projects we found either depend on costly per-slot sensors or still require guards to enter data by hand. No one before us had built a complete camera-based system that uses YOLOv8, Deep SORT, slot polygons, and a web dashboard specifically for a parking site in Bacolod City. That missing piece is the gap we set out to fill. SmartPark is our attempt to turn idle CCTV infrastructure into a working parking monitoring tool that is practical, affordable, and easy to scale for other sites. This project is not a continuation of previous work but a new integration of existing technologies tailored to the local setting, and it is needed because drivers waste time, guards struggle without real-time data, and no low-cost solution exists for Bacolod's parking problems.